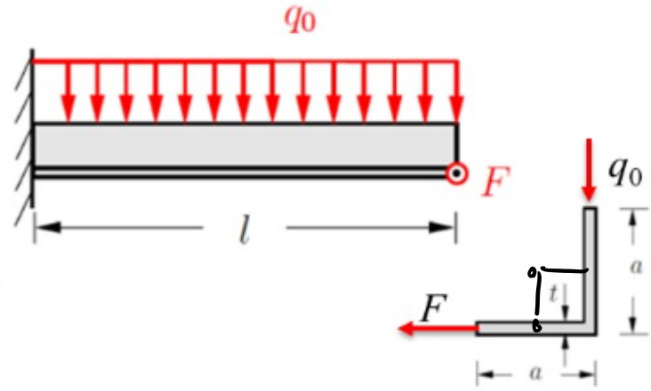


Homework 4.13 A cantilever beam with thin-walled profile ($t \ll a$) is subjected to a constant line load q_0 and a concentrated force F .

Determine the distribution of the normal stress in the cross section at the support.

Given: $F = 2q_0l$.



$$z_c = \frac{at(\frac{q}{4})}{2at} = \frac{q}{4} \quad \text{I} \quad \text{II}$$

$$I_{yI} = \frac{ta^3}{12} + (\frac{q}{4})^2(at)$$

$$I_{yII} = \frac{ta^3}{12} + (\frac{q}{4})^2(at)$$

$$I_y = I_{yI} + I_{yII} = \frac{ta^3}{12} + \frac{a^3t}{12} + \frac{a^3t}{12} \\ = a^3t \left(\frac{1}{12} + \frac{1}{12} + \frac{1}{12} \right) \\ = \frac{5}{12} a^3t = I_z$$

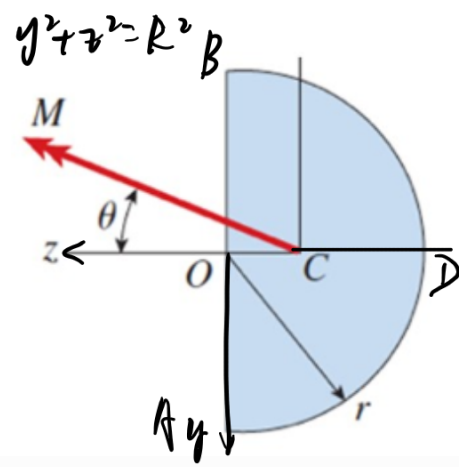
$$I_{yz} = -(\frac{q}{4})^2(2t)(l) = -\frac{q^2t}{8}$$

$$\Delta = I_y I_z - I_{yz}^2 = \left(\frac{5}{12} \right)^2 a^6 t^2 - \frac{a^6 t^2}{64} \\ = \frac{1}{36} t^2 a^6$$

$$M_y = -\frac{q_0 l^2}{2} \quad M_z = 2q_0 l^2$$

$$\sigma = \frac{1}{\Delta} \{ [M_y I_z - M_z I_{yz}] z - [M_z I_y - M_y I_{yz}] y \} \\ = \frac{36}{t^2 a^6} \left\{ \left[-\frac{q_0 l^2}{2} \frac{5}{12} t a^3 - 2q_0 l^2 \left(-\frac{t a^3}{8} \right) \right] z - \left[2q_0 l^2 \frac{5}{12} t a^3 + \frac{q_0 l^2}{2} \left(-\frac{t a^3}{8} \right) \right] y \right\} \\ = \frac{3}{4} \frac{q_0 l^2}{t a^3} (-17y + 7z)$$

Homework 4.14 A beam of semicircular cross section of radius r is subjected to a bending moment M having its vector at an angle θ to the z axis. Determine (a) formulas for the tensile stress σ_t and (b) the maximum compressive stress σ_c in the beam for $\theta=0$, 45° and 90° in the form $\alpha M/r^3$, with α as a function of θ .



$$a) \quad z_c = \frac{-\int \frac{z}{R} z dy}{\frac{1}{2}\pi R^2} = \frac{-\int z^2 dy}{\pi R^2} = \frac{-\int_0^R \frac{R^2 - y^2}{R^2} dy}{\pi R^2} = \frac{-2 \left[R^2 y - \frac{y^3}{3} \right]_0^R}{\pi R^2} = -\frac{4R}{3\pi}$$

$$I_z = \frac{1}{8}\pi R^4 \quad I_y = \frac{R^4}{72\pi} (9\pi^2 - 64) \quad I_{yz} = 0$$

at $\theta = 0^\circ$

$$\sigma_t = \sigma_A \quad \frac{M x_r}{I_z} = \frac{8M}{\pi R^3} \quad \sigma_t = \frac{8}{\pi} \frac{M}{R^3}$$

$$\sigma_c = \sigma_B = -\sigma_A = -\frac{8}{\pi} \frac{M}{R^3}$$

at $\theta = 90^\circ$

$$\sigma_t = \sigma_D \quad \sigma_t = \frac{M c}{I_y} = \frac{\frac{4R}{3\pi} M}{R^4 (9\pi^2 - 64)} \quad (72\pi)$$

$$= \frac{96M}{R^3 (9\pi^2 - 64)} = \frac{96}{9\pi^2 - 64} \frac{M}{R^3}$$

$$\sigma_c = \sigma_B = \frac{M(R-c)}{I_y} = \frac{(R - \frac{4R}{3\pi}) M}{R^4 (9\pi^2 - 64)} \quad (72\pi)$$

$$= \frac{(1 - \frac{4}{3\pi}) 72M}{R^3 (9\pi^2 - 64)} \approx 1.669 \frac{M}{R^3}$$